Name: _____ Class: _____ Physics Date: _____

Kinematics (Unit 2) Test

Achieved Points	Possible Points	% Total
	20	

TEST INSTRUCTIONS

1. **PRINT** your name and group at the top of this page.
2. Part 1 has 10 multiple-choice questions. **Carefully read each question and bubble your answer on the answer sheet—bubble only one answer per question.** Erase any errors wholly and carefully. Only answers on the bubble sheet will be graded.
3. Part 2 has 1 free-response question(s). **Answer all free-response parts on this document.**
4. Attempt ALL questions in the allotted time.
5. You will have 30 minutes to complete this exam.

ANSWER SHEET

1	A	B	C	D	6	A	B	C	D
2	A	B	C	D	7	A	B	C	D
3	A	B	C	D	8	A	B	C	D
4	A	B	C	D	9	A	B	C	D
5	A	B	C	D	10	A	B	C	D

You may remove this cover page. Formula sheet is on the back of this page .

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Grade 6 Formulas

$\bar{v} = \frac{\Delta x}{t}$	$s = \frac{d}{t}$	$\Delta x = x - x_0$	$a = \frac{\Delta v}{t} = \frac{v - v_0}{t}$
F_{net}	$F_g = mg$		
$W = Fd$	$P = \frac{W}{t}$		
$U = mgh$	$K = \frac{1}{2}mv^2$	$E = K + U$	
$I = \frac{V}{R}$			
$g = 9.8$ m/s^2	$e = 1.6 \times 10^{-19}$ C		

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Multiple Choice [10]

1. An athlete runs 300 m east on a straight road, then turns around and runs 100 m west. Which statement about the athlete's motion is correct?

   A. The total distance is 200 m and the displacement is 400 m east.

   B. The total distance is 400 m and the displacement is 200 m east.

   C. The total distance is 400 m and the displacement is 400 m east.

   D. The total distance is 200 m and the displacement is 200 m west.

2. A scout walks 1.4 km from camp to a lookout, then 0.9 km to a creek, and finally 0.7 km back to camp. How far did the scout walk in total?

   A. 0.7 km

   B. 3.0 km

   C. 2.3 km

   D. 1.4 km

3. A student measures how far a marble rolls along the floor and how far it ends up from where it started. Which SI unit should be used for both measurements?

   A. ampere (A)

   B. meter (m)

   C. kilogram (kg)

   D. second (s)

4. A research submarine is at rest 60 m below a floating marker. Using the marker as a reference, what is the submarine's position?

   A. level with the marker

   B. 120 m below the marker

   C. 60 m below the marker

   D. 60 m above the marker

5. A maglev train moves along a straight track. Measured from a signal post, the train's position changes from 15 m to 4 m. Find the train's displacement.

   A. +19 m

   B. +11 m

   C. −19 m

   D. −11 m

6. A tour guide tells the group, "We will meet at the fountain." The visitors use the fountain to describe where they are standing. What is the fountain acting as?

   A. a direction of motion

   B. a speed

   C. a reference point

   D. a displacement

7. A weather report says a storm is moving at 25 km/h toward the northeast. Which quantity is the report describing?

   A. the storm's speed

   B. the storm's distance

   C. the storm's velocity

   D. the storm's displacement

8. A passenger walks 1.5 m/s toward the front of a train while the train moves 9.0 m/s forward on a straight track. What is the passenger's velocity relative to the ground?

   A. 7.5 m/s forward

   B. 9.0 m/s forward

   C. 10.5 m/s backward

   D. 10.5 m/s forward

9. A rider on a Ferris wheel moves at a constant 2 m/s around in a circle. Which description of the rider is correct?

   A. The velocity is zero because the rider comes back to where they started.

   B. The rider's speed is constant, but the velocity keeps changing because the direction keeps changing.

   C. Both the speed and the velocity are constant.

   D. The speed is changing, but the velocity is constant.

10. A skateboarder rolling east begins to slow down. Taking east as the positive direction, the skateboarder's acceleration is ___.

   A. positive, because the skateboarder is moving east

   B. zero, because the skateboarder is still moving

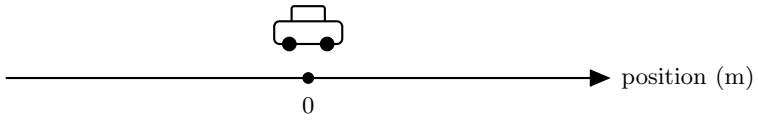
   C. positive, because the skateboarder is slowing down

   D. negative, because the acceleration points opposite the velocity

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Free Response [10]

11. A car travels along a straight road. The car first moves 90 m east in 10 s. It then moves 60 m west in 30 s. Finally, it moves 60 m east in 20 s.



(a) [2] Find the car's **total distance** travelled.

(b) [2] Find the **average speed** of the car over the whole journey.

(c) [2] Find the **total displacement** of the car over the whole journey. Give the magnitude and direction.

(d) [2] Find the **average velocity** of the car over the whole journey. Give the magnitude and direction.

(e) [2] Explain the difference between **adding scalars** and **adding vectors**. Use the car's trip as an example.